Practical Data Science

Mathematics of AI - RPTU Kaiserslautern

WiSe 25/26

Topics of the lecture

- Metrics
  - Classification and regression
- Handling a data science task
  - Understand the dataset and the task
  - Preprocess the data
  - Fit models and select the best
  - Presenting results

Metrics

Notebook `0_metrics.ipynb` contains a demonstration of how to compute performance metrics in various cases, including:

- Classification (binary and multiclass)
  - Confusion matrix, accuracy, AUC (with plots)
- Regression
- Clustering

Datasets

The notebook `1_download_datasets.ipynb` illustrates how to:

- get synthetic Datasets
- get simple benchmarks
- download datasets from OpenML

Preprocessing

The notebook `2_preprocessing.ipynb` guides through the preprocessing of a dataset, including:

- check for invalid values
- check for outliers
- scaling

Comparing models

A basic example of (nested) crossvalidation can be found in `3_crossvalidation.ipynb`, and the `4_comparing_models.ipynb` notebook illustrates how to use crossvalidation on a dataset to compare the performance of different models.

A few words...

This is the last Hands-on session...

Thank you for participating!

The next hands-on session slots will be devoted to Q&A for the data presentations (upon scheduling).